Activity 3.1 – RV32 ALU Extension

1.Design Approach:

Designed a 32-bit RISC-V Arithmetic Logic Unit (ALU) using Verilog HDL. Implemented arithmetic, logical, comparison and shift operations with operation selection using a control signal.

2.Verification Steps:

- Created a separate testbench.
- Applied different operand combinations.
- Tested addition and subtraction.
- Verified AND, OR and XOR operations.
- Tested shift operations.
- Checked comparison and LUI operations.
- Simulated using ModelSim.

3.Results:

The testbench checks ALU operation results for different control signals. PASS/FAIL messages are displayed.

4.Issues Faced:

Signed and unsigned operations must be handled correctly. Shift amounts must be restricted to the required bit width. ALU control signals must select the correct operation.

5.Conclusion:

Successfully designed and verified a 32-bit RISC-V ALU supporting multiple arithmetic and logical operations using Verilog HDL.